

Evaluation Harness: Accuracy, Constraints, and Generalization Probes

We measure three things the task requires, and report the cases where the model fails alongside the cases where it succeeds. It runs both the JEPA and the baseline through an identical pipeline so their numbers are directly comparable.

1. Predictive accuracy on held-out trajectories

We predict twelve months from twelve months of history for 250 patients whose seeds are disjoint from the training and validation sets, so no patient leaks across splits. We report overall mean squared error, per-field error, and error against forecast horizon. The baseline is more accurate overall.

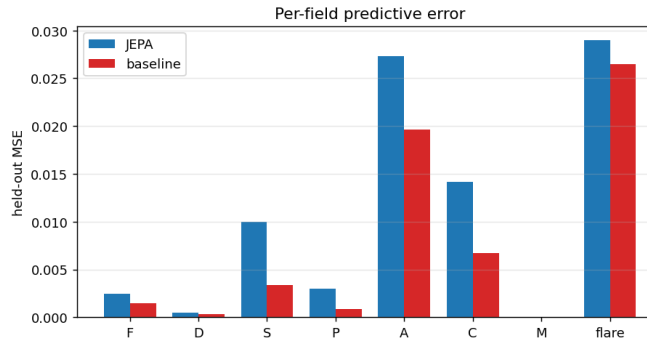


Figure 1 Held-out MSE vs Per-field predictive error

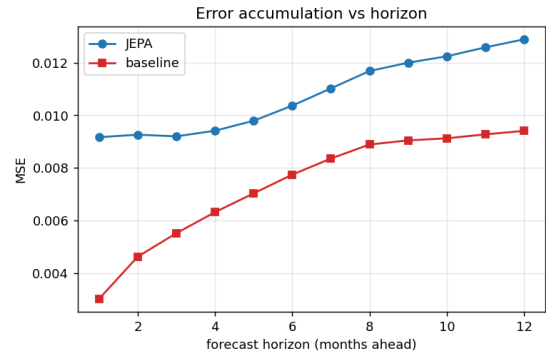


Figure 2 Error Accumulation vs Horizon

Metric	JEPA	Baseline
Overall MSE (12-step)	0.0106	0.0074
Cumulative error (sum over horizon)	0.128	0.088

Table 1: Held-out accuracy. Lower is better.

Per-field MSE	F	D	S	P	A	C	M	flare
JEPA	0.0025	0.0005	0.0100	0.0030	0.0273	0.0142	0.0000	0.0290
Baseline	0.0015	0.0003	0.0034	0.0009	0.0197	0.0067	0.0000	0.0265

Table 2: Per-field MSE.

The slow ratchet fields are easy; the fast fields (A, C, flare) carry most of the error because they are the stochastic part of the process. M is near zero for both because its increment is fixed by construction, which is a property of the parameterization and not evidence of learning.

Error grows with horizon. Predicting thirty months from twelve, the mean per-step error rises from 0.0089 within the training horizon to 0.0155 beyond it, a factor of about 1.75. This growth is inherent to autoregressive rollout.

2. Constraint-violation rate

A ratchet field that decreases, or a stricture that drops outside an ERCP month, or any field outside its bounds, counts as a violation. Because the guarantee is meant to hold for arbitrary parameters, we first test the constraint layer on 1,560 states generated from random inputs and network outputs of standard deviation five: zero violations. On the held-out predictions of the trained models, both the JEPA and the baseline report a violation rate of exactly 0.000 percent, which confirms the guarantee is a property of the architecture rather

than of one trained model. The rate remains zero over the thirty-month rollout, so accuracy and validity decouple: the model becomes less accurate far into the future but never becomes physically impossible.

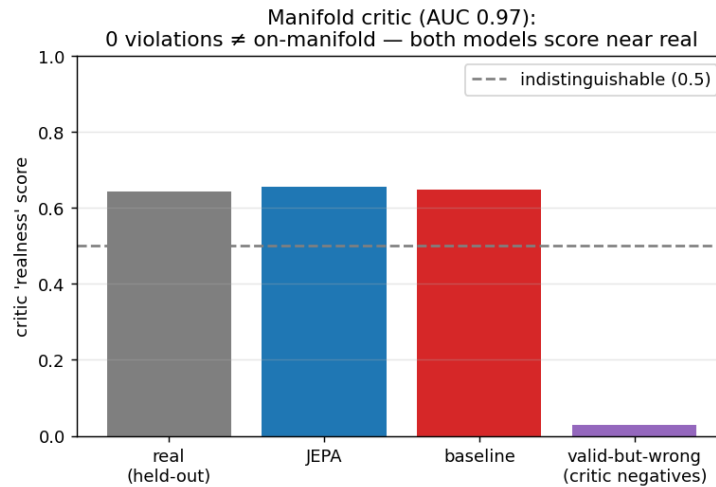


Figure 3 Manifold Critic

A zero violation rate is necessary but not sufficient, since a constraint-valid trajectory can still be dynamically wrong. We therefore train a discriminator to separate real generator transitions from constraint-valid but mis-scaled ones (AUC 0.97). Both models score at the real-data level (JEPA 0.66, baseline 0.65, real 0.64, against 0.03 for the valid-but-wrong negatives), so the predictions are on the generator's manifold and not merely legal.

3. Generalization probes, with failures shown

Each probe re-seeds the generator into a controlled cohort so that the shift being tested is the only thing that changes, and each states what it can and cannot establish. Two of the four are designed to break the model, and they do.

Probe	Result	Reading
Held-out susceptibility	high/low MSE ratio 1.33	Under-predicts rapid progressors; baseline degrades equally, so it is a data-support limit, not an architecture flaw.
Unseen treatment timing	MSE 0.0132 to 0.0145; S moves +0.013	The model is more sensitive to when the stricture-relieving ERCP lands than to UDCA timing. Small overall shift is mildly reassuring about a referral shortcut.
Long rollout (30 from 12)	error x1.75, violations 0	Accuracy decays; constraints hold. Accuracy and validity decouple.
Interpolation vs extrapolation	unseen/seen MSE 1.12	Small gap, so the model recovered the rule rather than the training set. Within this generator only.

Table 3: Generalization probes

We add two probes beyond the minimum. The noise-robustness probe compares the baseline and the JEPA denoised anchor across observation noise: the baseline error rises from 0.036 at sigma 0 to 0.106 at sigma 0.20, while the denoised anchor stays at 0.082 and overtakes the baseline near sigma 0.15. This locates the

area where the latent helps. The counterfactual probe starts UDCA six months earlier and validates against a generator re-run with a shared noise stream on responders: the model recovers the direction of the effect on all affected patients but underestimates the magnitude by about three orders (predicted -0.0001 against a true -0.0445). It knows the sign of the intervention and not its size, which is the expected limit of an observational model.

4. Conclusion

The baseline beats the JEPA on clean accuracy, including where we expected the latent to win. Both models satisfy every hard constraint by construction and both stay on the generator's manifold. The latent earns its place only under observation noise, through the denoised anchor. The model under-predicts fast progressors and recovers the direction but not the magnitude of a treatment counterfactual. The central limitation is that all of this is measured inside one generator, so the evaluation ranks models and locates failures but cannot settle whether the model is a world model or a generator inverter.